18 Cauchy's 1815 Theorem

Cauchy's 1815 theorem says that if a function of n variables takes more than two distinct values when you permute the variables, then it must take at least p distinct values, where p is the largest prime less than or equal to n. The number of distinct values always divides n factorial. Symmetric functions take one value, and the only way to get exactly two are by use of the alternating functions, like the Vandermonde product.

Theorem 18.1 (Cauchy's theorem:) *For a function of n variables, $f = f(x_1, \ldots, x_n)$, the number of distinct values under all permutations of the variables is either 1, 2, or at least p, where p is the largest prime less than or equal to n.*

Proof Let f be a function of n (distinct) variables $f = f(x_1, \ldots, x_n)$. Permuting the variables produces up to $n!$ possible inputs, but we assume f to take only R distinct values under these permutations. By Lagrange's theorem $n!$ is divisible R. Assume for contradiction purposes that $2 < R < p$. Consider a p-cycle σ (a cyclic permutation of order p on p of the variables, fixing the rest). The sequence of function values obtained by applying successive powers of σ to a fixed initial ntuple of variables:

$$f(x),\ f(\sigma(x)),\ f(\sigma^2(x)), \ldots,\ f(\sigma^{p-1}(x)).$$

There are at most p distinct values in the above list (since σ^p is the identity). Let

$$v_k = f(\sigma^k(x)) \text{ for } k = 0, 1, ..., p-1,$$

stand for these p values and lest us apply the **Pigeonhole Principle** (i.e., if you have more pigeons than pigeonholes, at least one hole has more than one pigeon). These p values are drawn from the set of R assumed distinct possible values that f can ever take, and we assume $R < p$. These values satisfy the cyclic relation: $v_{k+1} = f(\sigma^{k+1}(x))$ is obtained from v_k by applying $\sigma \pmod p$. By the pigeonhole principle at least two of these p values must be equal.

$$v_r = v_s \text{ for some } 0 \le s < r \le p-1.$$

Let $d = r - s$ (so $1 \le d \le p-1$).

Now apply powers of σ repeatedly. Because σ shifts the indices cyclically (i.e., applying σ moves v_k to $v_{(k+1) \pmod p}$), the assumption $v_r = v_s$ means the sequence is periodic with a smaller step. The equality propagates under shifts: Apply σ any number of times to both sides of $v_r = v_s$. This gives: $v_{r+m} = v_{s+m \pmod p}$ for any integer m. In other words, shifting the indices by the same amount preserves the equality. The index differences are multiples of d: The indices where we know equalities hold are exactly the index positions that differ by multiples of $d \pmod p$. That is, for any integer m: $v_k = v_{(k+m-d) \pmod p}$, whenever the equality chain reaches there.

Primality of p makes d a generator: Crucially because p is prime and d is between 1 and $p-1$, we have $\gcd(d, p) = 1$. Therefore, d has a multiplicative inverse modulo p, and the multiples

$$\{0 \cdot d, 1 \cdot d, 2 \cdot d, \ldots, (p-1) \cdot d\} \pmod p$$

hit every residue class modulo p exactly once and we cycle through all residues modulo p. Starting from any fixed index k, we can reach any other index l by adding a suitable multiple of d modulo p. Therefore $v_k = v_l$ for all k, l. In group theory terms: the subgroup generated by d modulo p is the entire cyclic group $\mathbb{Z}/p\mathbb{Z}$. Therefore entire orbit of $f(\sigma^k x)$ for $k = 0, 1, 2, \ldots, p-1$ collapses to a single value.

Conclusion:

The shifts by multiples of d generate the entire cyclic group of order p. It follows that all the v_k are equal (the orbit collapses to a single value). In particular, $v_0 = v_1$, i.e., $f(x) = f(\sigma(x))$. Since the initial ntuple x was arbitrary, this holds for every starting assignment: σ leaves f invariant (does not change its value). By choosing different p-cycles (or by conjugacy/relabeling of variables), this applies to all p-cycles in S_n.

This implies that all p-cycles leave f invariant (fix every value). One can then construct a 3-cycle as a product (or commutator) of two such p-cycles. Thus, 3-cycles also fix values. A 3-cycle is a product of two transpositions. Analysis of transpositions then shows that either f is fully symmetric ($R = 1$) or it takes exactly two values (like the sign/alternating case, e.g., Vandermonde product). But we assumed $R > 2$, a contradiction. You can't get three or more without hitting at least p values. ∎

Thus, no such R with $2 < R < p$ is possible. The pigeonhole step forces p-cycles (and then smaller cycles) to preserve values, collapsing the possibilities to $R = 1$ or 2. This is the heart of Cauchy's 1815 argument (sometimes phrased in terms of"forms" or "arrangement" of the variables). Modern presentations often frame it in terms of the action of the symmetric group S_n on the values, with stabilizers and orbits (see the **Orbit-Stabilizer Theorem**) but the core pigeonhole idea on powers of a cycle remains the same.

Why Primality of p Is Essential

If p were composite, say d shares a common factor with p, the multiples of $d \pmod p$ would only generate a proper subgroup, and you might only force equality within cosets rather than full invariance under σ. The primality ensures the generated subgroup is the whole group, forcing full invariance. This is the precise bridge from the pigeonhole collision to invariance under the full p-cycle. Once you have invariance under all p-cycles, the rest of the proof proceeds by constructing smaller cycles (e.g., 3-cycles) as products/commutators and analyzing transpositions.